METRIC

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10 June 2021
SUPERSEDING
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27 December 2010

FEDERAL SPECIFICATION

ACETYLENE, TECHNICAL, DISSOLVED

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two technical grades of acetylene dissolved in acetone.

1.2 Classification. The acetylene shall be of the following grades, as specified (see 6.2):

Grade A - 99.6 percent acetylene
Grade B - 98.0 percent acetylene

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FEDERAL STANDARDS

FED-STD-123 - Marking for Shipment (Civil Agencies)

COMMERCIAL ITEM DESCRIPTIONS

A-A-59860 - Valves, Cylinder, Gas (for Compressed or Liquefied Gases)

Comments, suggestions, or questions on this document should be addressed to Defense Logistics Agency Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or emailed to STDZNMGT@dla.mil. Since contact information can change, you may want toverify the accuracy of the address information using the ASSIST online database at https://assist.dla.mil.

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MILITARY SPECIFICATIONS

MIL-DTL-3701	- Cylinders, Compressed Gas: DOT-8 and DOT-8AL Acetylene, with Valves

MILITARY STANDARDS

MIL-STD-101	- Color Code for Pipelines and for Compressed Gas Cylinders
MIL-STD-129	- Military Marking for Shipment and Storage
MIL-STD-1411	- Inspection and Maintenance of Compressed Gas Cylinders
MIL-STD-1916	- DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil.)

CODE OF FEDERAL REGULATIONS

Title 49, Parts 171-185	- Hazardous Materials Regulations

(Copies of this document are available online at https://www.gpo.gov/fdsys/browse/collectionCfr.action?collectionCode=CFR.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ H1331	- Zero Acceptance Number Sampling Plans

(Copies of this document are available from https://www.asq.org.)

ASTM INTERNATIONAL

ASTM D329	- Standard Specification for Acetone

(Copies of this document are available from http://www.astm.org/.)

COMPRESSED GAS ASSOCIATION (CGA)

CGA C-13	- Guidelines for Periodic Visual Inspection and Requalification of Acetylene Cylinders
CGA G-1	- Acetylene
CGA G-1.1	- Commodity Specification for Acetylene

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CGA G-1.9	- Recommended Practices for Maintaining the Proper Solvent Level in Acetylene Cylinders
CGA V-9	- Compressed Gas Association Standard for Compressed Gas Cylinder Valves

(Copies of these documents are available from http://www.cganet.com/.)

3. REQUIREMENTS

3.1 Material. The acetylene shall conform to the requirements of gaseous acetylene as specified in CGA G-1.1 and shown in table I.

TABLE I. Properties of acetylene.

Limiting characteristics [1]	Grade A (Percent by volume) [2]	Grade B (Percent by volume) [2]
CGA G-1.1 Quality Verification Level	QVL H	QVL C
Acetylene ($C_2 H_2$)	99.6 minimum	98.0 minimum
Phosphine (PH_3) and arsine (AsH_3) [3]	0.0025 maximum	0.05 maximum
Hydrogen sulfide (H_2S) [3]	0.0025 maximum	0.05 maximum

[1] Additional information on the properties and handling of acetylene may be found in CGA G-1.
[2] Acetone-free basis.
[3] Carbide process acetylene only (see 6.3).

3.2 Cylinders and valves. The acetylene shall be contained in government-owned and furnished cylinders conforming to MIL-DTL-3701 equipped with valves meeting all requirements of A-A-59860. When specified (see 6.2), new cylinders meeting the same requirements shall be purchased and furnished with the gas product. When specified (see 6.2), the acetylene shall be contained in supplier-owned DOT-approved cylinders meeting the requirements of 49 CFR 171-185 equipped with valves qualified to either CGA V-9 or A-A-59860.

3.3 Cylinder maintenance. Government-owned and furnished cylinders shall be inspected, maintained, and when necessary, reconditioned in accordance with MIL-STD-1411 to meet all serviceability requirements of 49 CFR 171-185. Supplier-owned cylinders shall be processed using CGA C-13, CGA G-1.9, and the supplier's own approved procedures to meet all serviceability requirements of 49 CFR 171-185.

3.3.1 Valves. Unless otherwise specified (see 6.2), replacement valves for any defective valves found in the government-owned and furnished cylinders shall be furnished by the supplier and shall be qualified to meet all requirements of A-A-59860. Valves in supplier-owned cylinders shall be qualified to meet all requirements of either A-A-59860 or CGA V-9.

3.3.2 Acetone. Acetone for the replenishment of government-owned and furnished cylinders shall conform to ASTM D329.

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3.4 Capacity. Cylinders shall be filled to volume capacity on the basis of weight (see 6.6). Cylinders shall be filled in compliance with 49 CFR 173.303; the pressure in the filled cylinder shall not exceed 1723 kPa (gauge) at 21.1 °C (250 pounds per square inch gauge (psig) at 70 °F).

3.5 Leakage. Cylinders and valves shall not leak after being filled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards. The supplier shall verify that the production facility will produce acetylene that complies with the percent by volume (v/v) acetylene purity requirements of 3.1 for the grade supplied. Current analytical records of acetylene production by the facility may be used to perform the verification. The analytical records shall be considered current if taken within 180 days preceding the date of contract or purchase order. When current analytical records are not available, or in case of process upset or equipment malfunction which affects the product purity, not less than one acetylene sample from the production facility shall be tested as specified in 4.4.2.1 prior to starting or continuing production.

4.2 Quality conformance inspection.

4.2.1 Examination. Each filled cylinder shall be examined as specified in 4.3.1. Presence of one or more defects shall be cause for rejection of the cylinder.

4.2.2 Tests.

4.2.2.1 Individual cylinder. Each filled cylinder shall be tested for leakage as specified in 4.3.2.3. Failure of the test shall be cause for rejection of the cylinder.

4.2.2.2 Samples (carbide process acetylene only). Samples selected from each lot of carbide process acetylene shall be tested for phosphine/arsine and hydrogen sulfide as specified in 4.3.2.2. Sampling shall be from filled cylinders and shall be conducted in accordance with MIL-STD-1916 at Verification Level IV, ASQ H1331 at Index Value 1.0, or other similar accept at zero (c=0) plan, unless otherwise specified in the contract (see 6.2). A lot shall be considered to be all cylinders filled from the same source during a consecutive 24-hour period.

4.3 Inspection procedure.

4.3.1 Examination. The filled acetylene cylinders shall be examined for the following defects:

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a. Cylinder or valve not as specified in 3.2.
b. Cylinder maintenance not as specified in 3.3.
c. Capacity not as specified in 3.4.

4.3.2 Tests.

4.3.2.1 Acetylene purity. (This test is required only for verification of the production facility as outlined in 4.1.1.) Compliance with the acetylene purity requirement of 3.1 shall be determined by gas chromatography or other appropriate instrumental method. Prior to the test, a few cubic feet of gas shall be discharged to purge the sampling lines. The minimum acetylene content shall be 99.6 percent (v/v) for grade A and 98.0 percent (v/v) for grade B. Otherwise, compliance with the minimum requirements has not been met and shall constitute failure of this test.

4.3.2.2 Phosphine/arsine and hydrogen sulfide (carbide process acetylene only). Compliance of carbide process acetylene with the phosphine/arsine and hydrogen sulfide requirements of 3.1 shall be determined qualitatively as specified in 4.3.2.2.1. In case of dispute, the acetylene shall be quantitatively analyzed as specified in 4.3.2.2.2. Prior to each test, a few cubic feet of gas shall be discharged to purge the sampling lines.

4.3.2.2.1 Qualitative determination. The qualitative determination for both grade A and grade B acetylene may be made with color detector tubes for phosphine/arsine and for hydrogen sulfide (see 6.7). If the indicated percentage of phosphine/arsine or of hydrogen sulfide is greater than that specified in 3.1, the acetylene shall have failed the test.

4.3.2.2.2 Quantitative analysis. The quantitative determination of phosphine/arsine and of hydrogen sulfide shall be by gas chromatography or other appropriate instrumental method. The maximum allowance for phosphine/arsine is 0.0025 percent (v/v) for grade A and 0.05 percent (v/v) for grade B. The maximum allowance for hydrogen sulfide is 0.0025 percent (v/v) for grade A and 0.05 percent (v/v) for grade B. Quantities in excess of these respective amounts shall constitute failure of this test. Analysis by gas chromatography may be performed in lieu of the method specified in 4.3.2.2.1, if gas chromatography is the supplier's preferred method of testing.

4.3.2.3 Leakage. After filling, each cylinder shall be tested for leakage by applying a soap solution to all portions of the valve, the junction of the valve and cylinder, and the safety device on the cylinder. Care shall be taken to ensure that the solution does not contaminate the valve outlet. Any evidence of gas leakage as indicated by bubbling of the soap solution shall constitute failure of this test.

4.4 Inspection of preparation for delivery. All cylinders shall be examined to determine if any markings are missing, illegible, incorrect, or incomplete. Any one defect shall be cause for rejection.

5. PACKAGING

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5.1 Packing. Shipment of the cylinders of acetylene shall conform to 49 CFR 171-185.

5.1.1 Palletization. When specified (see 6.2), the cylinders of a single grade of acetylene shall be palletized together in the vertical, valve-up position, on hardwood pallets in accordance with the requirements of the procuring agency or activity.

5.2 Marking. The cylinders shall be marked in accordance with 49 CFR 171-185. Each cylinder shall be stenciled or tagged grade A or grade B as applicable.

5.2.1 Civil agencies. In addition to the marking specified in 5.2, the cylinders shall be marked in accordance with FED-STD-123.

5.2.2 Military agencies. In addition to the markings specified in 5.2, the cylinders shall be marked in accordance with MIL-STD-129.

6. NOTES

INFORMATION FOR GUIDANCE ONLY. (This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. The grade A acetylene covered by this specification is used as an illumination gas for navigational lights. The grade B acetylene covered by this specification is for use in cutting and welding.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Grade of acetylene required (see 1.2).
c. When supplier is to furnish cylinders (see 3.2).
d. When acetylene is to be contained in supplier-owned cylinders (see 3.2).
e. When government-furnished replacement valves are to be utilized (see 3.3.2).
f. Verification Level or Index Values if different from 4.2.2.2.
g. When cylinders are to be palletized (see 5.1.1).
h. Packaging requirements (see 5.1 through 5.2.2).

6.3 Processes. Acetylene is commercially produced by either of two types of processes:

a. Carbide process. Acetylene is produced by reaction of calcium carbide and water. The calcium carbide has varying amounts of phosphorus and sulfur contaminants, which result in phosphine, arsine, and hydrogen sulfide impurities in product acetylene.

b. Petrochemical processes. Acetylene is produced by the thermal or arc cracking of hydrocarbons, such as the Wulff process or the BASF process; or by the partial combustion of methane with oxygen, such as the Montecatini process. Acetylene so produced is sometimes referred to as petrochemical or by-product acetylene. There is no phosphine, arsine, or hydrogen sulfide contamination in acetylene produced by petrochemical processes.

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6.3.1 Grades. Acetylene produced by either type of process is considered to meet commercial cutting and welding requirements, which is the intended use of grade B, 98 percent (v/v) acetylene. Further purification produces grade A, 99.6 percent (v/v) acetylene, which is often described as purified grade. Generally, only carbide process acetylene is further purified.

6.4 Government-furnished cylinders. The contracting officer should arrange to furnish the cylinders specified in 3.2.

6.5 Maintenance of government-furnished cylinders. Purchasers should specify the extent to which the gas supplier should process government-furnished cylinders that require maintenance.

6.5.1 Service practices. The gas supplier should furnish at no additional cost all services that are required at each and every filling of a cylinder to comply with applicable regulations and normal good practice. Such services would include, but not be limited to all inspection, testing, evacuation, and handling services required for the gas supplied.

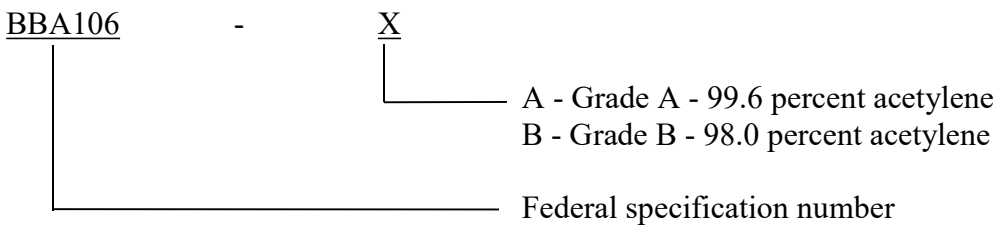
6.5.2 Service fees. A schedule of allowable fees should be specified by the purchaser for the gas supplier's performance of services such as the replacement of valves, valve parts, cylinder caps, periodic inspection, cleaning, painting, color-coding, marking, and handling of unserviceable cylinders as required. The gas supplier should furnish all materials and components for these services.

6.6 Basis of purchase. The basis of purchase should be the cubic foot. A conversion factor of 917.7 liters per kilogram (14.7 cubic feet per pound) weight of acetylene should be used. The weight of acetylene supplied in each cylinder is the difference between the filled weight and the tare (unfilled) weight of the cylinder. Acetylene is available in several cylinder sizes ranging from 283 through 11,893 liters (10 through 420 ft^3) rated capacity. In general, the amount of acetylene in a filled cylinder may vary slightly (up to 3 percent) from the rated volume capacity of the cylinder.

6.7 Color detector tubes. The color detector tubes specified in 4.3.2.2.1 are tubes containing granules impregnated with indicating chemicals. Two color detector tubes would be used for the specified tests, one for phosphine/arsine and one for hydrogen sulfide. A detector tube that does not have cross-sensitivity to hydrogen sulfide must be used when detecting for phosphine/arsine. A detector tube that does not have cross-sensitivity to arsine and phosphine must be used when detecting hydrogen sulfide. Phosphine/arsine color detector tubes cannot differentiate between phosphine and arsine content. The length of color change in a tube, when a measured volume of acetylene is passed through it, is directly proportional to the concentration of the impurity. The tubes and associated apparatus are available from several manufacturers and in several ranges.

6.8 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.

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6.9 Subject term (key word) listing.

Acetone
Arsine
BASF process
Carbide process
Cylinder
Gas chromatography
Hydrogen sulfide
Montecatini process
Petrochemical process
Phosphine
Valve
Wulff process

6.10 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

MILITARY INTERESTS:

Custodians:
Air Force - 68
DLA - GS

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FAS

Preparing Activity:
DLA - GS3

(Project 6830-2021-002)

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